Document 3 — Dataset Acquisition Guide

SIH26166 · Exactly what to download, from where, and in what order

Updated August 31, 2026 — merges Document 7 (Kaggle Training Workflow, Elaborated Setup & Dataset Checklist) into this document. Document 7 is now retired as a standalone file; its checklist and dataset-size guidance now live in Sections 2 and 5 below.

What changed in this update: Document 7 turned each of the five portals and two supporting datasets below into a literal, numbered checklist of the exact file/product to click, and stated plainly which pipeline stage actually consumes each one. That checklist is now Section 2's per-step reference table, and a new Section 5 states the 20 GB Kaggle dataset-size discipline that governs what may ever leave your machine as a Kaggle Dataset upload for training (see Document 1, Part A for the training side of this).

Contents

1. The five sources you actually need, and nothing else

Your PS names these sources directly. Don't go looking for alternatives — every one of these has a specific, non-interchangeable role.

#	Source	Role	URL
1	Chandrayaan Map Browse	Browse/footprint search for your source images	chmapbrowse.issdc.gov.in
2	ISSDC PRADAN	Full-resolution, PDS4-labelled raw source archive	pradan.issdc.gov.in
3	LROC NAC	Primary reference imagery	lroc.im-ldi.com
4	LROC QuickMap	Interactive coverage/coordinate check for the reference	quickmap.lroc.im-ldi.com
5	SELENE (Kaguya), JAXA	Secondary reference imagery	JAXA lunar data archive — search “SELENE Kaguya data archive”; JAXA's portal structure changes, use the current data-archive index rather than a bookmarked deep link

Plus two supporting datasets you'll need but that aren't named as portals in the PS:

#	Source	Role
6	A lunar DEM (LOLA, or the combined SLDEM2015 LOLA+Kaguya-TC mosaic)	Ground-truth elevation for synthetic sun-angle pair generation, ray–surface intersection, and Fourier-surface terrain rendering

#	Source	Role
7	A lunar crater catalog (e.g. the LPI/USGS Lunar Crater Database, or Robbins 2019 global crater catalog)	External, geometry-based anchor points for the self-consistency validation check (Log 04, Open 5)

2. Step-by-step: registering and pulling data, in the order to actually do it

Step 1 — Register on PRADAN, today, before anything else

This is explicitly the longest lead-time item in your own workplan.

- Go to pradan.issdc.gov.in.
- Create an account with institutional email if available (approval can be faster/more reliable for .ac.in/university-affiliated addresses).
- Fill the registration form; note that approval is manual and not instant — this is why it goes first, on day one, owned by one team member who does nothing else that day.
- While waiting, proceed to Step 2 using only the browse portal (no registration required for browsing).

Step 2 — Find a candidate scene pair on Chandrayaan Map Browse

- Go to chmapbrowse.issdc.gov.in.
- Use the footprint/region search to browse OHRC, TMC-2, and IIRS coverage. Prioritize a well-documented, named crater (easier to cross-check visually and against a crater catalog later) rather than an arbitrary coordinate.
- Note down: instrument, orbit/scene ID, approximate lat/lon bounding box, acquisition date (needed later for sun-angle/ephemeris computation).
- Pick 2–3 candidates, not just one — some will fail the coverage check in Step 3.

Step 3 — Confirm reference coverage on LROC QuickMap before committing

- Go to quickmap.lroc.im-ldi.com.
- Enter the same lat/lon bounding box from Step 2.
- Confirm NAC imagery actually exists over that footprint at a usable resolution/incidence angle — do this before downloading anything full-resolution. OHRC's footprint is tiny (~12×3 km), so a near-miss in coordinates can mean zero actual overlap.
- If NAC coverage is thin or absent, check SELENE coverage for the same region as a fallback — treat it as a genuinely separate case, not a substitute with identical properties (different resolution, different sun-angle history).
- Repeat for your 2–3 candidates; keep the one with the cleanest confirmed overlap.

Step 4 — Pull the full-resolution source data via PRADAN (once approved)

- Log in to pradan.issdc.gov.in.
- Search by the orbit/scene ID noted in Step 2.
- Download the PDS4-labelled product: an .xml label file paired with an .img raw data file (OHRC/TMC-2), or a hyperspectral cube for IIRS.
- Verify the download against the label's stated checksum/size before assuming it's good — a truncated download of a binary .img file will not necessarily throw an obvious error when opened.

Step 5 — Pull the reference product

- From LROC NAC (lroc.im-ldi.com): locate the exact NAC product ID confirmed in Step 3, download it (also a PDS-family format).

- If using SELENE: locate the equivalent product on JAXA's archive for the same footprint.

Step 6 — Get a DEM for your region

- LOLA gridded data products (NASA PDS Geosciences Node) give a lunar-wide altimetry DEM at various resolutions (e.g. 118 m/px global, higher-resolution regional mosaics where available).
- SLDEM2015 (a combined LOLA + Kaguya Terrain Camera DEM, higher resolution than LOLA alone in the ±60° latitude band) is the better choice if your chosen crater falls in that band — check the PDS Geosciences Node's SLDEM2015 archive page for coverage and download instructions.
- Crop to your scene's footprint immediately after download; don't carry a lunar-wide DEM through your whole pipeline.

Step 7 — Get the crater catalog subset for your region

- Pull the relevant regional subset from the LPI/USGS Lunar Crater Database or the Robbins (2019) global lunar crater catalog (search the current hosting location — typically distributed as shapefiles/CSV via the USGS Astrogeology Science Center or the PDS Annex).
- Filter to craters whose rims fall inside your chosen scene's footprint; these become your external, geometry-based validation anchors (Log 04, Open 5).

Step 8 — Verify the ISRO SAC benchmark citation before it goes anywhere near a slide

- The pitch's central “receipt” is the 2025 ISRO Space Applications Centre paper (Makharia et al.) benchmarking classical vs. deep-learning feature matching on Chandrayaan-2 imagery.
- Verify the exact title, author list, venue, and DOI yourself before citing it — a fabricated-sounding-plausible citation is a real risk category, and this is the one citation every judge in the room is most likely to ask you to produce on the spot.

2.1 Exact checklist — every source, the exact item to pull, and where it lands

Section 1 already names the five portals and two supporting datasets. Below is that same list collapsed into a literal checklist of what file(s) to click, so there's no ambiguity about which product on each portal you're pulling, and which pipeline stage actually consumes each one.

#	Source	Exact item to pull	Where it lands
1	PRADAN (pradan.issdc.gov.in)	PDS4-labelled product for your chosen orbit/scene ID: the paired .xml label + .img raw array (OHRC/TMC-2) or hyperspectral cube (IIRS)	data/raw/<instrument>/<scene_id>/
2	Chandrayaan Map Browse (chmapbrowse.issdc.gov.in)	No download here — used only to note the orbit/scene ID, lat/lon bounding box, and acquisition date that you then search for on the other portals	notes only
3	LROC NAC (lroc.im-ldi.com)	The specific NAC product ID confirmed against your footprint on QuickMap — a PDS-family raster	data/reference/lro_nac/<product_id>/
4	LROC QuickMap (quickmap.lroc.im-ldi.com)	No download — coverage/coordinate confirmation only, before pulling anything full-resolution	notes only
5	JAXA SELENE archive	Fallback reference product for the same footprint, only if NAC coverage is thin/absent. Search the current data-archive index (portal structure changes)	data/reference/selene/<product_id>/

#	Source	Exact item to pull	Where it lands
6	LOLA gridded data (PDS Geosciences Node) or SLDEM2015	A DEM tile covering your scene's footprint — SLDEM2015 if your crater falls within ±60° latitude (higher resolution), else LOLA	data/dem/<region_name>_sldem2015.tif
7	LPI/USGS Lunar Crater Database or Robbins (2019) catalog	Regional CSV/shapefile subset filtered to craters whose rims fall inside your scene's footprint	data/crater_catalog/<region_name>_craters.csv

Order to actually do this in (unchanged, restated for convenience): Step 1 register on PRADAN first (longest lead time) → Step 2 browse candidates on Chandrayaan Map Browse → Step 3 confirm coverage on QuickMap → Step 4 pull the PRADAN source once approved → Step 5 pull the LROC/SELENE reference → Step 6 pull/crop the DEM → Step 7 pull/filter the crater catalog → Step 8 verify the Makharia et al. 2025 ISRO SAC citation before it goes near a slide.

What actually feeds the descriptor training workflow (Document 1, Part A): only the DEM tile from Step 6 is consumed there — cropped into patches, then rendered into synthetic sun-angle pairs. The raw OHRC/TMC-2/NAC imagery from Steps 1, 4, and 5 is used later, in the FastAPI `/match` pipeline (Document 4), not in descriptor training itself.

3. File formats you will actually open, and with what

Format	Instrument	Read with
PDS4 .xml label + .img raw array	OHRC, TMC-2	rasterio, GDAL, or pds4_tools
Hyperspectral cube (explicit BSQ/BIP/BIL band order)	IIRS	spectral package, or manual numpy reads respecting the label's stated interleave
PDS-family raster	LRO NAC	rasterio/GDAL (LRO products are widely GDAL-compatible)
PDS-family raster	SELENE/Kaguya	rasterio/GDAL; verify projection/datum metadata separately (Log 07 geodesy gap)
GeoTIFF/IMG DEM	LOLA / SLDEM2015	rasterio
Shapefile/CSV	Crater catalog	geopandas / pandas

None of these open in a standard image viewer. Build your ingestion layer against GDAL/rasterio from day one, generically across all three source instruments, per your own workplan's Log 02 checklist.

4. A ready-to-run data folder structure

```
data/
|-- raw/
|   |-- ohrc/<scene_id>/{*.xml,*.img}
|   |-- tmc2/<scene_id>/{*.xml,*.img}
|   `-- iirs/<scene_id>/{*.xml,*.cube}
|-- reference/
|   |-- lro_nac/<product_id>/*
|   `-- selene/<product_id>/*
|-- dem/
|   `-- <region_name>_sldem2015.tif
|-- crater_catalog/
|   `-- <region_name>_craters.csv
`-- processed/
    |-- synthetic_pairs/
    `-- metrics/
```

Keep raw/ and reference/ read-only once downloaded (checksum-verified); do all cropping/reprojection/synthetic-pair generation into processed/, never mutating the archive originals in place.

5. Keeping every Kaggle upload under the 20 GB cap

This section governs what may ever leave your machine as a Kaggle Dataset upload once you reach the training workflow in Document 1, Part A. Kaggle enforces a 20 GB cap per private dataset (and a separate 20 GB cap on a notebook's own /kaggle/working output at commit time). Nothing acquired via Sections 1–2 above needs to come anywhere near that limit if the rule below is followed — the actual bottleneck is almost always someone uploading a full scene instead of a patch.

Artifact (from this document's sources)	Typical size	Ever uploaded to Kaggle?
Raw PRADAN OHRC/TMC-2 .img scene (Step 4)	1–8 GB per scene	Never — stays local/PRADAN-side; only crops leave
LRO NAC / SELENE reference product (Step 5)	Hundreds of MB–few GB	No — consumed later by the FastAPI /match pipeline (Document 4), not by training
Cropped DEM patch, from Step 6's tile (128–512 px, GeoTIFF)	0.1–5 MB each	Yes — this is what goes in the lunar-dem-patches-v1 dataset
Crater catalog subset, from Step 7 (CSV/shapefile)	A few MB regionally	Only if a stretch-goal notebook needs it as a mounted input; filter to footprint first

The only thing that should ever be produced *from* a Kaggle notebook and pushed back up as a new dataset is the trained model export (`descriptor.onnx`, 1–5 MB) described in Document 1, §A.4.7–A.4.8. Full mechanics, the six-cell training script, and the path-selection rationale live there; this section only states the acquisition-side discipline that keeps every upload small in the first place: crop to patches immediately after download (Steps 6–7 above), and never zip or upload a raw PRADAN/LROC/SELENE product as a Kaggle Dataset.